



MTH205: ANALYSIS IN ONE VARIABLE

MidSem1 (Spring Semester 2025), IISER Mohali

SET E

Timing: 12 pm to 1 pm

Date: February 04, 2024

Max. Marks: 20

Name:

Reg. No:

1. Please answer all questions. The answer must have a justification/explanation otherwise, zero marks will be given even though the answer is correct.

2. Your name and registration number should be written on the question paper and answer sheet. At the time of submission, the question paper must be attached.

1. Find all points of discontinuity for the following functions and give a correct justification for your answer.

(a) $\sec(x)$ [2]

Solution: $\sec(x) = \frac{1}{\cos(x)}$ defined on $\mathbb{R} \setminus \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$. This follows that the points of discontinuity of $\sec(x)$ is $\{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$.

(b) $h(x) = \frac{1}{x} \sin(x)$ for $x \in \mathbb{R} \setminus \{0\}$ and $h(0) = 1$. [2]

Solution: We use the fact that $|\sin(\theta)| \leq |\theta|$ and $|\cos(\theta)| \leq 1$, for any $\theta \in \mathbb{R}$ to show that $\sin(x)$ is a continuous function on $\mathbb{R}$. Let $a \in \mathbb{R} \setminus \{0\}$ then

$$\begin{aligned} |\sin(x) - \sin(a)| &= \left| \sin\left(\frac{x+a}{2} + \frac{x-a}{2}\right) - \sin\left(\frac{x+a}{2} - \frac{x-a}{2}\right) \right| \\ &= 2 \left| \cos\left(\frac{x+a}{2}\right) \sin\left(\frac{x-a}{2}\right) \right| \\ &\leq 2|x-a| \end{aligned}$$

For any $\varepsilon > 0$ choose $\delta = \varepsilon$ such that $|\sin(x) - \sin(a)| < \varepsilon$ whenever $|x-a| < \delta$. This implies that $\frac{\sin(x)}{x}$ is continuous for every $x \in \mathbb{R} \setminus \{0\}$.

On the other hand, if there is a sequence (in $\mathbb{R} \setminus \{0\}$) $\{x_n\} \rightarrow 0$ as $n \rightarrow \infty$ then

$$\lim_{n \rightarrow \infty} h(x_n) = \lim_{n \rightarrow \infty} \frac{\sin(x_n)}{x_n} = 1 = h(0).$$

That is, h is continuous at 0. Hence there are not points of discontinuity for $h(x)$.

(c) $g(x) = x - \lfloor x \rfloor$, for all $x \in \mathbb{R}$. [2]

Solution: Let $x \in \mathbb{Z}$. Then $g(x) = 0$ and

$$\begin{aligned} g\left(x - \frac{1}{n}\right) &= \left(x - \frac{1}{n}\right) - \left\lfloor x - \frac{1}{n} \right\rfloor \\ &= \left(x - \frac{1}{n}\right) - (x-1) \end{aligned}$$

$$= 1 - \frac{1}{n}.$$

It follows that $\{x - \frac{1}{n}\} \rightarrow 0$ but $\{g(x - \frac{1}{n})\} \rightarrow 1 \neq g(x)$. Thus g is not continuous at every $x \in \mathbb{Z}$. Hence the points of discontinuity for g is, $\mathbb{Z}$.

- (d) Let $A = \{x \in \mathbb{R} : x > 0\}$ and $f: A \rightarrow \mathbb{R}$ be defined by [4]

$$f(x) = \begin{cases} 0, & \text{if } x \in A \cap \mathbb{R} \setminus \mathbb{Q} \\ \frac{1}{n}, & \text{if } x = \frac{m}{n} \in A, \gcd(m, n) = 1. \end{cases}$$

Solution: If $x \in \mathbb{Q}$ and $x > 0$ then there is a sequence $\{x_n\}$ of irrational positive numbers such that $\{x_n\} \rightarrow x$ as $n \rightarrow \infty$. From the definition, $f(x_n) = 0$ for $n \geq 1$ and $f(x) > 0$. This means that $\{f(x_n)\}$ does not converge to $f(x)$. Thus f is discontinuous at every rational number $x \in A$.

Now consider $x > 0$ is an irrational number and $\varepsilon > 0$ then by Archimedean property there exists $N \in \mathbb{N}$ such that $\frac{1}{N} < \varepsilon$.

Suppose there is a rational number $\frac{p}{N} \in (x - 1, x + 1)$ then there are finitely many such choices for $p \in \mathbb{Z}$ and hence we can choose a small $\delta > 0$ such that $(x - \delta, x + \delta)$ contains no rational numbers with denominator less than N .

It follows that for $y \in A$ with $|y - x| < \delta$ we have

$$|f(y) - f(x)| = |f(x)| \leq \frac{1}{N} < \varepsilon.$$

Thus f is discontinuous at every irrational number $x \in A$. Therefore, the points of discontinuity for f is $A \cap \mathbb{R} \setminus \mathbb{Q}$.

2. Give one example with a justification, for each of the following.

- (a) The functions f, g are defined on $A \subseteq \mathbb{R}$ such that both f, g are discontinuous at some point $x_0 \in A$ but $f + g$ and $f \cdot g$ are continuous at x_0 . [2]

Solution: Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}; \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Now we see that if $x \in \mathbb{Q}$ there is a sequence $\{x_n\}$ in $\mathbb{R} \setminus \mathbb{Q}$ such that $\{x_n\} \rightarrow x$, however $f(x_n) = 0$ for every n and so, $\{f(x_n)\}$ does not converge to $f(x) = 1$. Similarly, if $y \in \mathbb{R} \setminus \mathbb{Q}$ then there is a sequence $\{y_n\}$ in $\mathbb{Q}$ such that $\{y_n\} \rightarrow y$, however $f(y_n) = 1$ for every n and so, $\{f(y_n)\}$ does not converge to $f(y) = 0$. Therefore, f is discontinuous at every $x \in \mathbb{R}$. On the other hand, $f + g = f \cdot g = 0$, which is a continuous function on $\mathbb{R}$.

- (b) A compact set in $\mathbb{R}$ which is nowhere dense. [2]

Solution: The Cantor Set is compact and nowhere dense in $\mathbb{R}$. The Cantor Set C is given by

$$C = \bigcap_{n=0}^{\infty} C_n,$$

where $C_0 = [0, 1]$ and C_n is the union of 2^n closed intervals of the form $[\frac{k}{3^n}, \frac{k+1}{3^n}]$. Therefore, being the intersection of closed subsets of $[0, 1]$, the Cantor set C is compact set in $\mathbb{R}$.

Suppose C contains a non-empty open interval $I = (a, b)$, then $(a, b) \subset C_n$ for each $n \in \mathbb{N}$. Since C_n is the union of 2^n closed intervals, each of length $\frac{1}{3^n}$ and the total length of the intervals (whose union is C_n) is $\left(\frac{2}{3}\right)^n$. It follows that $0 \leq b - a \leq \left(\frac{2}{3}\right)^n \rightarrow 0$ as $n \rightarrow \infty$. Hence I is empty set and that shows that the Cantor set contains non-empty open interval. Equivalently, C is nowhere dense set.

- (c) A continuous function $f: A \rightarrow \mathbb{R}$ such that $\{f(x_n)\}$ is not necessarily Cauchy in $\mathbb{R}$ whenever $\{x_n\}$ in A is Cauchy. [2]

Solution: Consider the function $f(x) = \frac{1}{x}$ on $(0, \infty)$. If $\{y_n\}$ is any convergent sequence in $(0, \infty)$ that converges to $y \in (0, \infty)$ then $\{f(y_n)\} \rightarrow f(y)$ as $n \rightarrow \infty$. Thus f is continuous function.

However, if $\{\frac{1}{n}\}$ is a Cauchy sequence in $(0, \infty)$ but $\{f(\frac{1}{n}) = n\}$ is not Cauchy sequence. This shows that a continuous function $f: A \rightarrow \mathbb{R}$ not necessarily maps Cauchy sequence to a Cauchy sequence.

3. Let $[a_1, b_1] \supset [a_2, b_2] \supset [a_3, b_3] \supset \cdots \supset [a_k, b_k] \supset \cdots$ be a decreasing sequence of non-empty closed interval in $\mathbb{R}$ with $a_n \neq b_n$ for every $n \geq 1$. Then show that

- (a) $\bigcap_{n=1}^{\infty} [a_n, b_n] \neq \emptyset$. [3]

Solution: The set $S = \{a_k : k \in \mathbb{N}\}$ is bounded above by b_1 and hence $x := \sup S$ exists in $\mathbb{R}$ by least upper bounded property. By choice $a_k \leq x$, for every $k \in \mathbb{N}$. Since each b_k is an upper bound for S , we also have $x \leq b_k$ for every $k \in \mathbb{N}$. Thus $x \in \bigcap_{n=1}^{\infty} [a_n, b_n]$

- (b) Further, if $\{b_n - a_n\} \rightarrow 0$ as $n \rightarrow \infty$ then $\bigcap_{n=1}^{\infty} [a_n, b_n]$ is a singleton set. [1]

Solution: Clearly, $x \in \bigcap_{n=1}^{\infty} [a_n, b_n]$. If there is $y \in \bigcap_{n=1}^{\infty} [a_n, b_n]$ then $|x - y| \leq b_n - a_n$ for every $n \in \mathbb{N}$. Since $\{b_n - a_n\} \rightarrow 0$, as $n \rightarrow \infty$ we get $x = y$.

*** ALL THE BEST ***